

Motivations

The purpose of abstraction

The purpose of abstraction is to unify more examples (not get further away from them). Sometimes category theory can feel like arbitrary abstractions for the sake of it; in that case I think something is missing. It's true that sometimes abstraction is driven by some internal logic, but if the resulting abstraction doesn't unify anything then I would be suspicious of it.

Structural understanding

I think that category theory is about understanding structural reasons behind things, not just constructing formal justifications. Some proofs in math just proceed step by step and arrive at the conclusion by stringing all those steps together. I think a truly categorical proof does more than that, uncovering some deep aspect of *why* something is happening, structurally.

One example was when we saw that the category of factors of 30 has the same shape as the category of factors of 42. We could show that those categories are isomorphic in several ways. Here they are, sort of in order of what I consider increasing structural depth.

1. Construct a functor from one to the other, and show that it's a bijection on objects and on morphisms.
2. Construct a functor from one to the other, and an inverse for it.
3. Show that each one is the category of factors of abc , where a, b, c are distinct primes.
4. Show that each one is a product category I^3 , where I is the quintessential arrow category.

I hope it is clear that none of these proofs is more correct than the others, and they are all expressed in category theory. But they become increasingly categorical in the sense of the deep categorical structure that they uncover.

“Economical” category theory

Working with category theory usually involves categories with specific kinds of structure in them, like products and coproducts, other limits and colimits, and so on. One approach is a sort of “rich” approach where you throw tons of structure into your category because you think you will always have that in any of the categories you're working in: “We might as well assume we have products because all the categories we work in will always have products.” It's the